# **Resting Electrocardiogram System**

Biomedical Instrumentation - Final Term Paper

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## ABSTRACT

This project presents a custom-designed biomedical instrumentation (BMI) system for high-fidelity ECG acquisition, emphasizing improved signal clarity through hardware-level conditioning. Scientifically, the system addresses persistent challenges in ECG diagnostics—namely, noise interference and poor signal quality—by implementing a tailored analog front end. Technically, the system features a three-lead electrode configuration coupled with an instrumentation amplifier (gain = 100), a 60 Hz notch filter for power-line noise suppression, and a second-order low-pass filter with a 150 Hz cutoff to eliminate muscle and ambient artifacts. Simulink-based validation confirmed the filtering stages preserved ECG morphology while attenuating interference. Broader applications of this platform include wearable cardiac monitoring, real-time clinical diagnostics, and integration into telehealth systems, offering a scalable foundation for future digital health technologies.

## KEYWORDS

Electrocardiogram (ECG), Instrumentation Amplifier, Notch Filter, Low-pass Filter, Simulink

## BACKGROUND

An electrocardiogram (ECG or EKG) is a non-invasive diagnostic tool used to record the heart's electrical activity over time. By detecting the electrical impulses generated during cardiac depolarization and translating them into waveform tracings, ECGs provide critical insights into heart rate, rhythm, and conduction patterns. This enables clinicians to detect a range of cardiac conditions, including arrhythmias, ischemic changes, and myocardial infarctions. The ECG is widely used in both emergency and routine clinical settings due to its speed, reliability, and ability to provide immediate results. Technological advancements have led to the development of portable and handheld ECG devices and the integration of artificial intelligence for enhanced diagnostic accuracy[1]. As cardiovascular disease remains the leading global cause of death, claiming approximately 17.9 million lives annually, the demand for ECG systems continues to grow. The global ECG market was valued at $8.62 billion in 2023 and is projected to reach $20.87 billion by 2034, driven by aging populations and rising rates of heart disease[2]. The electrocardiogram (ECG) is a critical tool in modern medicine, offering a fast and accurate method for assessing the heart's electrical activity using surface electrodes. It is commonly used to evaluate cardiac rhythm, rate, and conduction abnormalities, helping clinicians identify life-threatening conditions early. Typical symptoms that may prompt an ECG include chest pain, dizziness, palpitations, shortness of breath, and fatigue. In the U.S. alone, cardiovascular disease causes one death every 33 seconds and accounts for one in every five deaths, underlining the importance of timely ECG access. The ECG's role in both prevention and early detection makes it a cornerstone in cardiovascular care and a continuously evolving technology in the medical device market.

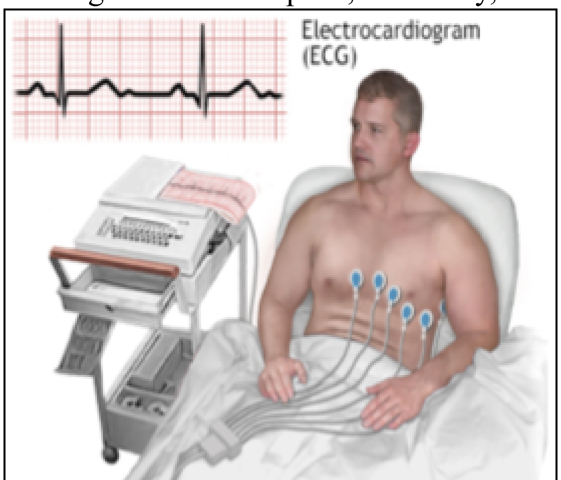


**Figure 1.** Electrocardiogram (ECG) Procedure: A patient undergoing an ECG, a widely used diagnostic test for detecting heart diseases. The image shows ECG electrodes attached to the patient's chest, connected to an ECG machine that records the heart's electrical activity[2].

## LITERATURE REVIEW AND RESEARCH GAPS

In 1903, Willem Einthoven revolutionized cardiac diagnostics by inventing the string galvanometer, making clinical ECG recording possible. Early ECG instruments were bulky and limited to hospital settings, but advances such as oscillographs and electrocardiographs made these devices more practical for everyday clinical use[3]. Consequently, the introduction of computerized ECG systems marked a major advancement, allowing for serial comparisons crucial for monitoring myocardial infarctions. While digital ECG systems have improved diagnostic capabilities, several key challenges remain. Signal quality is often compromised by baseline wander and high-frequency noise, reducing diagnostic accuracy. Lead misplacement and user error can distort waveforms, leading to misinterpretation. Additionally, ECG data is often difficult to integrate with modern Electronic Health Record (EHR) systems, creating barriers to streamlined clinical workflows. Patient compliance also presents an obstacle, as discomfort from electrode patches can discourage routine testing and long-term monitoring.[4–5]

## MOTIVATION AND OBJECTIVES

Accurate ECG signal acquisition is routinely degraded by noise, interference, and poor signal conditioning, lowering diagnostic effectiveness. Baseline wander, 60 Hz power-line noise, and high-frequency artifacts may obscure significant waveform features, especially in low-amplitude signals. This project is motivated by the need to improve hardware-level signal quality before digital processing. The main objectives are to: (1) design a high-gain amplification circuit capable of boosting weak biopotentials; (2) integrate a 60 Hz notch filter to suppress electrical interference; (3) implement a 150 Hz low-pass filter to reduce high-frequency noise while preserving diagnostic waveform components; and (4) ensure the system enables clean, interpretable ECG signals for improved downstream analysis and clinical utility.[6]

## TECHNICAL HIGHLIGHTS

### *1) BMI System Composition*

**Fig. 2** depicts the key components of a 3-lead ECG system. A core technical element of this BMI system is the ECG signal acquisition and processing chain, consisting of surface electrodes, signal conditioning, and digital processing. A standard 3-lead configuration is used, with electrodes on the right arm (RA), left arm (LA), and left leg (LL) to capture the heart's electrical activity. These biopotentials are first routed into an instrumentation amplifier, which provides high common-mode rejection and amplifies low-amplitude ECG signals. While typical ECG applications use a gain >1000, a gain of 100 is used here for safety—to limit amplification of potential transients or voltage spikes that could saturate the system or pose a risk during testing.[7-9]

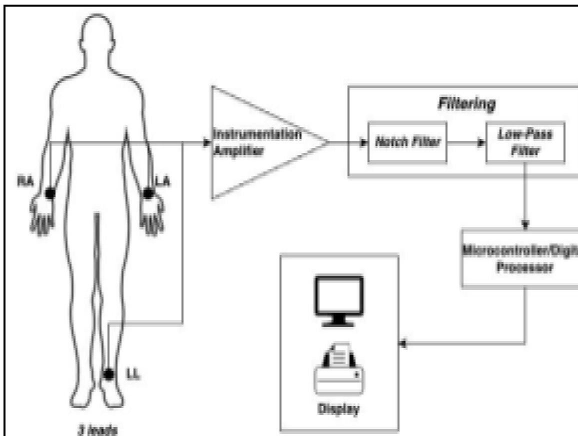


**Figure 2.** Schematic of a basic ECG-based BMI system. The signal is acquired via a 3-lead electrode configuration, amplified using an instrumentation amplifier, then filtered through a notch and low-pass filter. The processed signal is digitized and analyzed by a microcontroller or digital processor, with output displayed on a monitor or printer.

The amplified signal then passes through a filtering stage, beginning with a notch filter to remove powerline interference. Standard ECG notch filters target 60 Hz with a narrow 5–10 Hz bandwidth to preserve important ECG components. This is followed by a low-pass filter to eliminate high-frequency noise while preserving key cardiac features. The low-pass filter targets noise sources such as muscle artifacts and electromagnetic interference. Muscle noise, or EMG interference, arises from skeletal muscle activity and overlaps with ECG frequencies, especially during involuntary movement or posture shifts. Additional noise during resting ECGs includes motion artifacts, respiration-induced drift, and ambient electronic interference.[7-8]

Following American Heart Association guidelines, a 150 Hz cutoff was used for diagnostic-quality ECGs. A second-order low-pass filter was chosen for sharper attenuation above cutoff, and the gain was set to 1 to maintain signal amplitude and avoid distortion. The filtered signal is then digitized and processed by a microcontroller or digital processor, which may extract features or classify signals. Finally, the processed output is displayed in real time on a monitor or printer. This signal path ensures clean, high-quality ECG signals are captured and analyzed, forming the basis for BMI system control or monitoring.[9]

***2) Pivotal circuit and associated validation***

As shown in **Fig. 3**, the resting ECG circuit design consists of three electrical elements. The INA is composed of three op-amps, in which the input signals are connected to the non-inverting op-amps, and the third op-amp is a differential amplifier which rejects noise and provides a cleaner signal amplification. The gain of the INA is calculated using the equation (1). The NF has three resistors (1.6kΩ x 2, and 425Ωk) and three capacitors (100nF x 2, and 1uF) with the aim of attenuating power line interferences at a signal frequency of 60 Hz. Finally, the second-order LPF has two resistors (12kΩ and 138kΩ) and two capacitors (10nF) with a cut-off frequency of 150 Hz. All op-amps in this circuit design are powered with +5V and -5V DC power supply and current of 0.1A.[10-11] The constructed full circuit board is shown in **Fig 4a**.

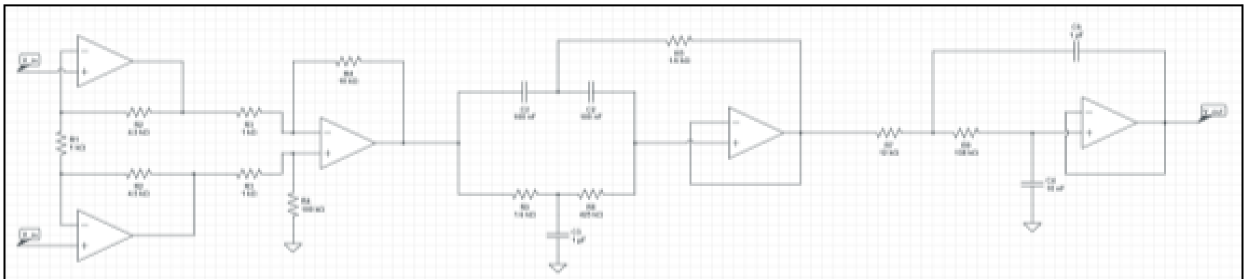

**Figure 3**:The pivotal circuit of the resting electrocardiogram system consists of, from left to right, an instrumentation amplifier (INA) with a gain of 100, a notch filter (NF), and a second-order low pass filter (LPF).

**<u>Hardware Testing</u>**

*<u>Instrumentation Amplifier</u>*

To test the functionality of the INA, simulated ECG input signals are provided from the waveform generator at a frequency of 1.2 Hz, which falls in the range of a typical bandwidth of an ECG signal, and amplitude of 2mVp. Due to the nature of small physiological ECG signals, a gain of 100 is used here instead due to safety concerns and only for verification purposes to ensure the component is functional. The resistor values were chosen arbitrarily in setting a gain equal to 100, as demonstrated in the **Eq. (1)**.

$$Gain = (1 + \frac{2(R_2)}{R_1})(\frac{R_4}{R_3}) = (1 + \frac{2(4.5k\Omega)}{1k\Omega})(\frac{10k\Omega}{1k\Omega}) = 100 \qquad (1)$$

The oscilloscope image shows the performance of our instrumentation amplifier (INA) element, but the results deviate from the expected operation (**Figure 4b**). The yellow trace shows a high frequency signal measured at 90 MHz, suggesting significant noise or interference is being picked up—possibly from environmental sources or power supply switching noise. The green trace, representing the output of the INA, shows a signal with a frequency of approximately 52.8 kHz and an amplitude of around 600 mV, which is far greater than the expected 200 mV output given a gain of 100. They indicate that the circuit may be affected by instability, or poor grounding. Due to external noise or instability, the INA might be amplifying unintended high frequency components for our application, but the constructed INA is functioning in terms of basic amplification. Further investigation and filtering would help clarify the output.[10-12]

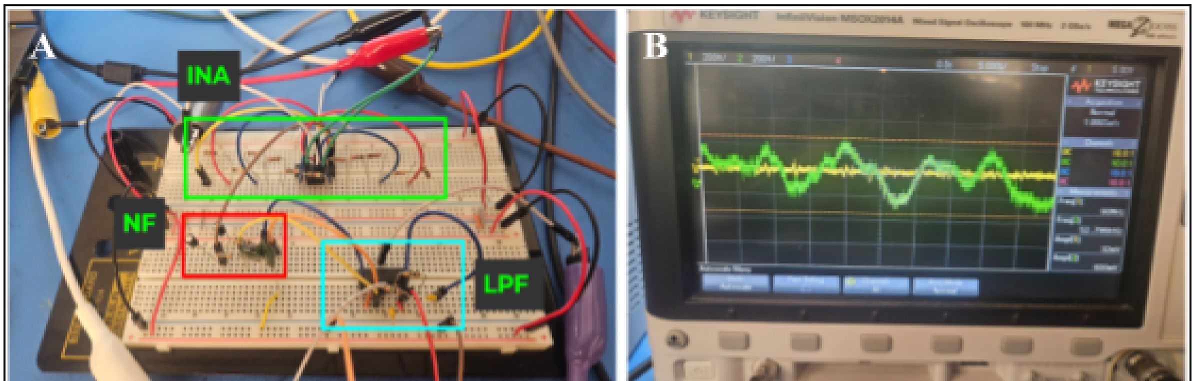


**Figure 4: A:** The full setup of the circuit on the breadboard. **B:** The oscilloscope output of the INA component of the circuit.

*Notch Filter*

Both input and output voltages were collected for input frequencies ranging 15-300 Hz and amplitude of 1V to create a NF frequency response curve (**Fig. 5a**). The notch filter frequency response curve shows a sharp attenuation centered at 60 Hz, with a dip in gain of around -3 to -3.5 dB, effectively targeting and reducing power line interference, which is a common source of noise in biomedical signal acquisition. This allows the filter to suppress the 60 Hz component without significantly affecting nearby frequencies, thus preserving the low-frequency ECG signal. The gain (dB) is obtained by dividing the output voltage over the input voltage, and the log of the value is multiplied by 20 for each measured input frequency.[10-12]

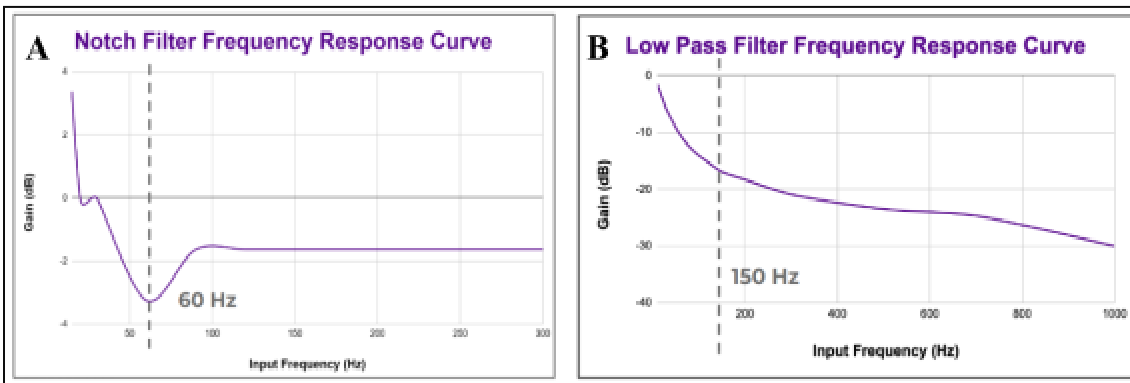


**Figure 5**: Frequency response curve of the **A:** NF component, and **B:** LPF component.

<u>*Low Pass Filter*</u>

Following NF validation, a similar procedure was performed to validate the working condition of LPF. Both input and output voltages were collected for input frequencies ranging 10-1000 Hz and amplitude of 1V to create a LPF frequency response curve (**Fig. 5b**). The low-pass filter frequency response curve shows a second-order roll-off beginning at 150 Hz, gradually attenuating higher-frequency components. This filter is designed to remove high-frequency noise and artifacts, such as those from muscle artifacts or electromagnetic interference, which are not part of the desired ECG signal.[10-12]

**<u>Software Modeling - Simulink</u>**

To validate the ECG signal processing pipeline, a Simulink model was built to simulate ECG generation, amplification, and filtering, including a 1000x gain stage, a 60 Hz notch filter, and a ~150 Hz low-pass filter. The model was run over a 10-second interval, and the output was visualized using a scope. As shown in the figures, the unfiltered signal (**Fig. 6a**) exhibited high-amplitude noise and baseline instability, while the filtered signal (**Fig. 6b**) displayed clean, well-defined ECG waveforms with reduced noise and sharper QRS complexes. This confirms that the implemented filters effectively attenuate unwanted high-frequency components, validating the software model's ability to replicate realistic ECG signal conditioning.

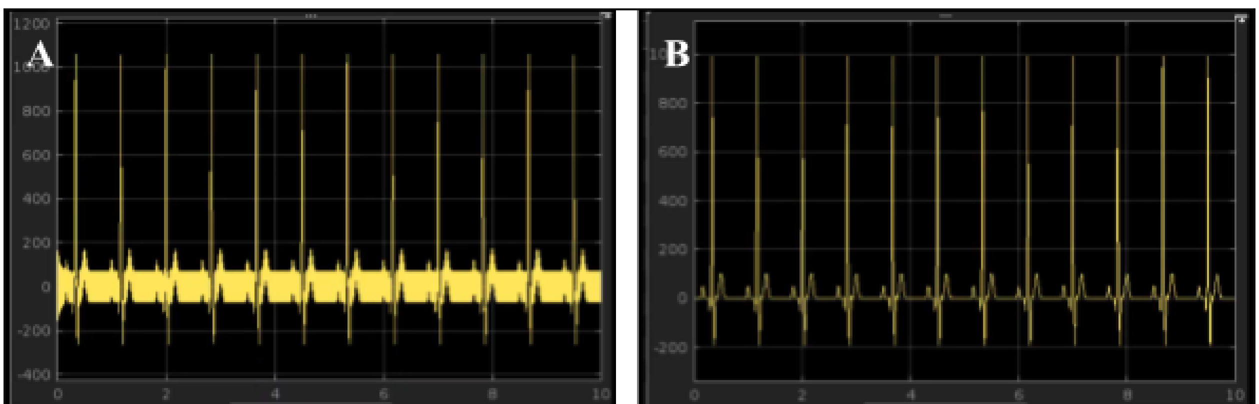


**Figure 6**. Comparison of simulated ECG output with and without filtering in Simulink. **A**: Unfiltered ECG signal immediately after amplification, exhibiting significant high-frequency noise and baseline drift. **B**: Filtered ECG signal after amplification, notch filtering (60 Hz), and low-pass filtering (150 Hz),

## CONCLUSION

The primary contribution of this BMI system is its ability to cleanly capture ECG signals using early-stage amplification and filtering, which reduces the need for heavy post-processing and supports real-time monitoring. By integrating signal conditioning including amplification, notch filtering, and low-pass filtering, this system addresses key challenges related to noise and signal clarity in resting ECG measurements. Broader applications of this system extend to remote cardiac monitoring, wearable health devices, and intelligent prosthetics. Improvements can be made by incorporating machine learning algorithms to offer promising directions for both technical advancement and expanded clinical impact.

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